

Cache Controller FSM Description

Controller Function

The controller manages cache hits, misses, writebacks, refills, and metadata updates.

FSM States

The controller contains IDLE, LOOKUP, HIT_READ, HIT_WRITE, MISS_DECIDE, WRITEBACK, REFILL, UPDATE_METADATA, and RESPOND states.

IDLE

Waits for a CPU request.

LOOKUP

Checks cache tag, valid bit, and determines hit or miss.

HIT_READ

Handles successful read operations from cache.

HIT_WRITE

Updates cache data on write hit and sets the dirty bit.

MISS_DECIDE

Checks whether the victim cache line is dirty.

WRITEBACK

Transfers dirty cache line data to main memory word by word.

REFILL

Loads requested cache line words from main memory.

UPDATE_METADATA

Updates tag, valid, and dirty bits after refill.

RESPOND

Signals transaction completion to the CPU.